

“Missing/Unspecified”: Demographic Data Visualization During the COVID-19 Pandemic

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Abstract

While data¹ has shown that COVID-19 disproportionately affects Black people, the CDC's early data listed race as “missing/unspecified” at high rates. Incomplete demographic data obscures the virus's full impact on marginalized communities. Without more information about who the virus is affecting and how, we cannot protect our most vulnerable. This article demonstrates disconnects between reported datasets and data visualizations in public-facing COVID health and science communication and suggests steps that technical and professional communicators can take in creating or using data visualizations accurately and ethically to describe COVID conditions and impacts.

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As Jones et al. (2016) argued, technical communication as a field has long been committed to social justice. Melonçon and Warner (2017) suggested that technical and professional communication (TPC) scholars can make an impact in the end products of data visualization with their expertise in user experience and audience analysis. Wolfe (2015) emphasized the importance of selecting and manipulating data before visualizations are created, arguing that selecting data to report without the appropriate context is unethical and, even if the data is accurate, misleading. Given how powerful data-driven arguments can be, we must ensure that when we create data visualizations or teach them to our students, we attend to their potential for advocacy and firmly root data stories in context (Loukissas, 2019).

Further, data is often communicated in visualizations that are crafted to tell a specific story, and context is often lost in the process of simplifying the data; similarly, data is often touted as objective, inherently factual information, untouched by human interference (“raw” data). As Gitelman (2013) reminded us, however, data is never “raw,” or free from interpretation and explanation, on any level; human actors always play a part in how data is created and communicated. As a result, rather than acting as an objective third party, data reifies those actors’ existing power structures and inequalities.

Data visualization in health communication is no exception. While data already shows that COVID-19 disproportionately affects Black people (Garg et al., 2020), as of April 14 data from the Centers for Disease Control and Prevention (CDC, 2020a) lists race as “missing/unspecified” in 78% of cases. Almost all of the tools created to help citizens track the virus’s magnitude, spread, and impact focus specifically on cases and deaths broken down by county, state, and country; the Johns Hopkins University dashboard (2020) is a particularly good example of this. Most of these tools allow users to look at the data on a map over time or to focus on a particular area, but almost all the available data does not show demographic information. Such incomplete demographic data obscures the virus’s full impact on marginalized communities.

In this study, I analyze two types of CDC data visualizations: (a) basic tables and graphs that are not labeled visualizations by the CDC and (b)

Demographic characteristics of COVID-19 cases in the United States, as of April 14, 2020*						
Age group (years)	No. of cases (% of total)					Total
	< 18	18-44	45-64	65+	Unknown	
Totals	7,001	145,499	147,003	90,619	8,730	398,852
Race missing/unspecified	5,958 (85%)	119,683 (82%)	115,997 (79%)	63,449 (70%)	5,687 (65%)	310,774 (78%)
Race specified	1,043 (15%)	25,816 (18%)	31,006 (21%)	27,170 (30%)	3,043 (35%)	88,078 (22%)
Among those with race specified						
American Indian or Alaska Native	2 (0.2%)	128 (0.5%)	106 (0.3%)	54 (0.2%)	3 (0.1%)	293 (0.3%)
Asian	41 (4%)	1,238 (5%)	1,368 (4%)	826 (3%)	217 (7%)	3,690 (4%)
Black or African American	301 (29%)	8,092 (31%)	10,666 (34%)	7,729 (28%)	374 (12%)	27,162 (31%)
Native Hawaiian or other Pacific Islander	4 (0.4%)	131 (0.5%)	114 (0.4%)	57 (0.2%)	10 (0.3%)	316 (0.4%)
White	683 (65%)	16,120 (62%)	18,662 (60%)	18,452 (68%)	2,438 (80%)	56,355 (64%)
Multiple/other	12 (1.2%)	107 (0.4%)	90 (0.3%)	52 (0.2%)	1 (0%)	262 (0.3%)
Missing/unspecified	5,962 (85%)	12,0883 (83%)	11,8912 (81%)	68,857 (76%)	4,191 (48%)	318,805 (80%)
Hispanic/Latino	315 (4%)	5911 (4%)	5108 (3%)	2097 (2%)	572 (7%)	14,003 (4%)
Non-Hispanic/Latino	724 (10%)	18,705 (13%)	22,983 (16%)	19,665 (22%)	3,967 (45%)	66,044 (17%)

*Case notifications were received by CDC from U.S. public health jurisdictions and the National Notifiable Diseases Surveillance System (NNDSS).

Figure 1. Screenshot of the CDC’s (2020a) April 14 demographic data.

argument-focused, storytelling graphs that are explicitly labeled “data visualizations” by the CDC. I compare these sources with the COVID Tracking Project’s (2020) COVID Racial Data Tracker, compiled and updated by *The Atlantic*’s journalists and the Antiracist Research and Policy Center, because at this time, it is one of the few data sources that explicitly focus on race in COVID cases as a central concern. Finally, I suggest steps that TPC practitioners, instructors, and students can take to accurately and ethically create or teach data visualizations.

CDC COVID-19 Tracking Tools and Visualizations

The CDC (2020a) published demographic data as of April 14, 2020, in a detailed table, breaking down total cases (both count and percentage) by race, ethnicity, and age, including the amount of missing or unspecified data (see Figure 1). While this table showed nationwide data rather than breaking it down by state or county, it still showed the level of uncertainty in the data; almost 80% of the cases reported had no racial or ethnicity information. So while the table clearly shows that Black and White people constituted the majority of cases with race specified, accounting for the 80% of unspecified cases could drastically change the picture.

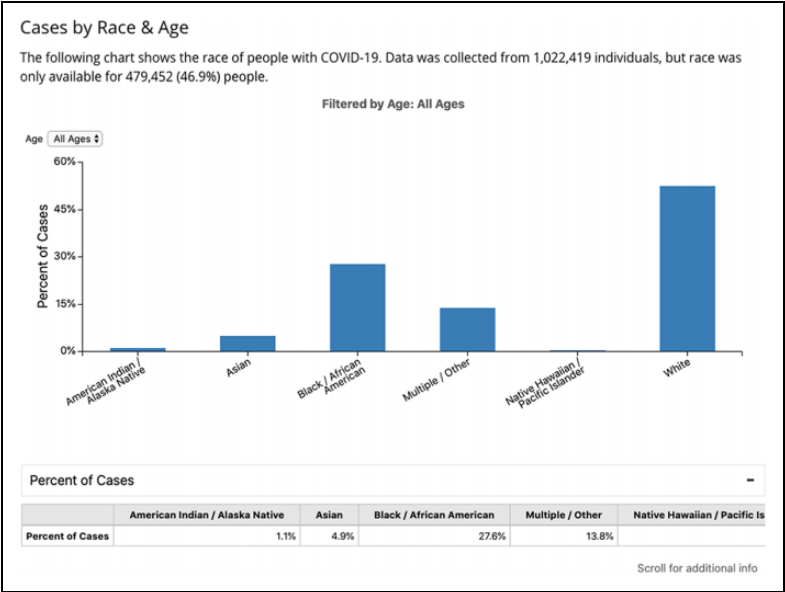


Figure 2. A screenshot of the CDC’s (2020a) COVID cases by race and age visualization (updated from the table in Figure 1 in May 2020).

By May 11, 2020, the CDC’s (2020a) data reporting strategy changed: The complex table from April was gone, replaced with a simple bar graph and single-row table representing percent of cases, filterable by age group (see Figure 2). Although the graph made it easier to visually compare case percentages across racial lines, its new filtering option showed far less data at one time and did not allow users to directly compare data from different age groups. Also, the April table clearly reported uncertainty in the data by including proportions of missing or unreported demographic data in the table itself whereas the May graph and table reported this information in the text above the visualization, making it easier to overlook. As in the April table, the data is nationwide rather than broken down or filterable by state, so users cannot obtain information from the data about their own communities or whether COVID’s impact on different racial groups is proportionate to their population size in affected communities.

Both the April and May efforts were intended to report COVID-19 cases as broken down by demographics; nothing on the tables, graph, or the surrounding Web page attempted to draw conclusions from or otherwise

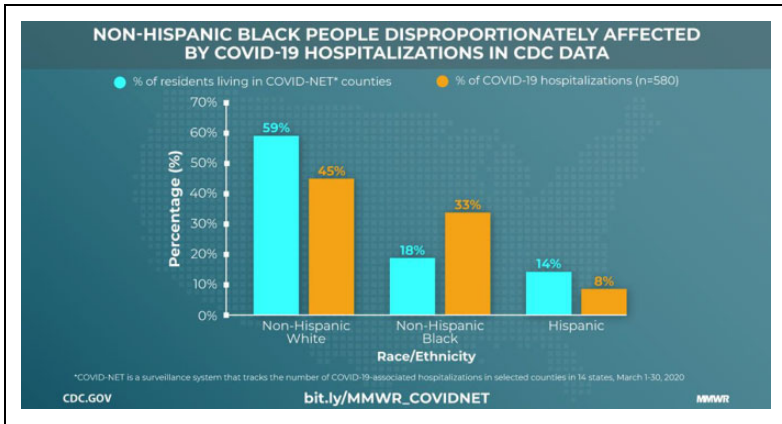


Figure 3. Screenshot of CDC’s (2020b) data visualization of COVID-19 hospitalizations, broken down by race/ethnicity.

interpret the data for viewers. The table also did not color code, shade, or otherwise highlight any particular variable or cell to draw viewers’ attention. The page was not labeled as a data visualization either time I examined it; I call it a tracking tool because that seems to be its intended purpose—to be a neutral tool showing facts without making a point. But as I indicated, from the initial data collection that did not focus on demographic data (almost 80% of cases had no associated racial data in April) to the choice to restrict data to nationwide or, in May, one age group at a time, these tools invite some questions and preclude others.

On a different page, however, the CDC has published explicitly labeled data visualizations that clearly focus on storytelling rather than reporting (2020b). Stacked bar and line graph visualizations are posted in both static image and animated GIF format, with explanatory headings and color differentiation that focus attention on the relevant information and let other information fade into the background. Each visualization makes an argument, both in textual form via explanatory titles and captions and in visual form via the data. One graph adds a race and ethnicity component in order to make the point that Black people are disproportionately hospitalized with COVID in the data (see Figure 3). These visualizations not only name insights easily gleaned from the data shown, but they also draw attention to these insights and add new contextual data to answer a particular question (e.g., the proportion information in Figure 3); rather than reporting, these visualizations interpret and explain data in ways that guide viewers to an

insight. While these visualizations tend to highlight how different groups, particularly marginalized ones, are affected by COVID-19, their national focus does not allow users to investigate the virus's impact on their own community. Broad claims are useful in certain contexts or for national organizations such as the CDC, but citizens interested in taking action locally need local data.

The Racial Data Tracker

The COVID Tracking Project's (2020) Racial Data Tracker (RDT), created explicitly to understand how and where COVID-19 is affecting people of color by comparing cases and deaths to state population percentages, is not technically a data visualization; the dashboard under development at the time of this writing is a spreadsheet that walks the line between the CDC's tracking tools and visualizations. Although the RDT's dashboard does not visually emphasize any particular data points, it compiles key variables together in a set (state, population, cases, deaths, race) to make it easy for users to ask questions about how people of color are being affected—such as why in Indiana, Black people constitute about 18% of the COVID-19 cases but only about 9% of the population. The RDT consciously foregrounds, in Wolfe's terms (2015), a particular interpretative level: state rather than nation, race rather than age.

Key Takeaways

Comparing the CDC's COVID-19 tracking tools (2020a) and data visualizations (2020b) reveals a stark contrast in strategy; tracking tools are meant to report while visualizations are meant to interpret and explain. But as TPC scholars, we know that in reporting information, we are always interpreting and explaining; ignoring that fact leads to thin, unhelpful visualizations such as the CDC's (2020a) tracking tools, which do not answer important questions about marginalized communities. We must ensure that our reports have enough context and depth that they will not be misleading. I present some key takeaways from these visualizations for TPC practitioners, teachers, and students.

Balance simplicity with context. A simple visualization might be easy to read, but its lack of context precludes users from asking many questions of the visualization. We could employ strategies such as adding variables by creating a stacked bar graph instead of a simple one or making the visualization a set of side-by-side graphs for ease of comparison. Folks in TPC

are already used to working with user-centered design, and these principles could work well in data visualization, as we can see when comparing the usefulness of the CDC's tracking tools and visualizations to that of the RDT.

Ask not only what questions visualizations can answer but also what they cannot answer. If the visualization's answers do not include questions about marginalized communities, we must ask why and consider if and how they can be expanded accordingly. Data is created, not found, so marginalized communities must be considered and included before the data is even collected; otherwise, we might not have inclusive data available to visualize, as in the absence of nearly 80% of the CDC's racial data (2020a).

Even those visualizations that only seek to report information cannot be neutral or without interpretation. Thus, TPC scholars, students, and practitioners must make sure that their data visualizations attend to context and to the needs of all users, including marginalized ones, regardless of how neutral they aim to be.

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Note

1. I use data as a mass noun with singular verbs in this article because I discuss data as a mass rather than a collection of individual data points.

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